

5(a)	allow one mark if monocyte <u>and</u> neutrophil both labelled as phagocytes and no other cell is labelled phagocyte allow phagocyte for either the monocyte or neutrophil if the other cell correct   red blood cell / erythrocyte ; monocyte / macrophage; I white blood cell I leucocyte   neutrophil ; A polymorphonuclear leucocyte / basophil / eosinophil lymphocyte ; I B / T I B-cell / T-cell	4
5(b)	15 / 15.0 / 15.1 / 15.2 (kPa) ; 0.09 / 0.10 / 0.1 (s) ;	2

Question	Answer	Marks
5(c)(i)	<i>emphysema and chronic bronchitis do not need to be named</i> <i>any one from must be in context of COPD</i> less, air / oxygen, able to reach, alveoli / air sacs / gas exchange surface ; less <u>oxygen</u> able to, diffuse to / AW, capillaries (from, alveoli / air sacs) ; ventilation impaired / AW ; <i>any two from effects of chronic bronchitis and emphysema</i> inflammation (of respiratory airways) ; build up / accumulation / excess / AW, of mucus decreased lumen of, trachea / bronchus / bronchioles / respiratory tubes / airways ; A airway obstruction I blocked A constriction / narrowing (loss of, elastin / elastic fibres, so) less ability (of alveoli) to recoil ; alveoli, burst / destroyed or alveolar walls break down ; large(r) air sacs (instead of alveoli) / AW ; decreased / reduced, SA:V ; <i>of, alveolar area / gas exchange surface area</i> AVP ; e.g. relevant effect of pulmonary hypertension affecting gas exchange	3
5(c)(ii)	<i>any two from</i> decreases quantity of, <u>oxygen</u> / O_2, taken into blood / absorbed / transported (back to heart) ; AW A fewer red blood cells to take up <u>oxygen</u> A blood less oxygenated decreases quantity of, <u>carbon dioxide</u> / CO_2, leaving blood / excreted / AW ; A more carbon dioxide remains in the blood (than normal) increased, ventilation / breathing, rate ; R breathes more deeply AVP ; e.g. further increase in pulmonary hypertension	2